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- By **Sim Shuzhen**

Facing Up To The Future

With facial recognition technology now widely used in law enforcement and in consumer devices, has privacy become a relic of the past?

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AsianScientist (Jan. 26, 2018) – Part of the allure of a big city is the anonymity it affords its denizens—the freedom, relished by many, of disappearing into a crowd and wandering unnoticed, if only for a few hours. Unnoticed, that is, by one's fellow human beings, whose brains are limited in their capacity for remembering the many details that make each individual unique.

These limits, however, do not apply to powerful computers running the latest facial recognition algorithms—with a big enough database, they can identify an individual from amidst a throng of people in a matter of seconds. In a simulated exercise, Chinese officials took a mere seven minutes to locate BBC reporter John Sudworth in the city of Guiyang, using a vast network of CCTV cameras paired with facial recognition software.

With such systems now being used for surveillance in cities all over the world, the notion of being just another face in the crowd is fast becoming obsolete.

Seeing holistically

To be used in real-time law enforcement, facial recognition software must do its job quickly and accurately, performing well even under non-ideal conditions, said Mr. Chris de Silva, head of global face recognition solutions at Japanese tech giant NEC.

> "In real life, we don't usually have a perfect view of a person, like one you would find in a passport photo. Someone might have a beard one day and not the next; they could be wearing sunglasses; or the lighting or image quality could be bad," said de Silva.

In general, facial recognition algorithms work by converting multiple points on your face into a numeric 'faceprint' or encoding. NEC's NeoFace algorithms handle difficult images, said de Silva, by looking more holistically at a face rather than at specific facial landmarks—better mimicking what humans do in real life.

The company's facial recognition technology has been deployed by police departments around the world, including in the city of Surat, India, where officials credit it with helping to reduce crime rates

by 27 percent.

> "By having good surveillance platforms in cities, you not only have the capability to respond to crime, but also to create an environment that deters criminals," said Mr. Walter Lee, evangelist and government relations leader at NEC's Global Safety Division. "Our goal is to reduce crime, not to catch more criminals."

Who owns your face?

In addition to law enforcement, facial recognition is also increasingly being used for access control—unlocking your Apple or Android smartphone, for instance. Facebook and Google use the technology to identify people (and even pets, in Google's case) in images; meanwhile, paying with your face has taken off in a big way in China, with companies like Face++ and SenseTime as the major players.

> "At this point, facial recognition is going to be heavily dominated by industry," said Associate Professor Cham Tat-Jen, who studies computer vision at Singapore's Nanyang Technological University. "Google, in particular, has access to a lot of data that is not publicly available, and this enables them to create a really accurate face recognizer. I don't think anyone in academia has that amount of data nor the computational resources to compete."

Cham himself used to work on face detection, but has since moved on to 3D telepresence, developing technologies aimed at allowing people thousands of kilometers apart to hold realistic conversations. Given the dominance of for-profit companies in the facial recognition space, how worried should the public be about privacy?

According to NEC's de Silva, 'Big Brother'-type scenarios are overblown. NEC, he says, does not store data; instead, it provides an empty application to law enforcement agencies, which the agencies then populate with their own databases of persons of interest. In realtime surveillance, a person's image data is dropped if it does not match anyone in this repository.

> "There is no big database in the sky or in the cloud," de Silva said.

Tipping points and tradeoffs

De Silva thinks that society will eventually reach a tipping point—the more the public learns about what facial recognition actually is and what it can do, the less afraid it will become of the technology.

> This is especially so in countries at risk of terrorist attacks. "People expect their governments to do whatever they can to protect them, using the tools that are available. So you have this kind of balance between how much privacy you want to give away versus how secure you want to be," he said.

For Cham, who is in favor of using facial recognition technology for law enforcement purposes, privacy is already an illusion.

> "I think we live in a world where we have to assume that we don't have much privacy, and so things like security become more important," he said.

And because hackers can now so easily impersonate someone else in cyberspace, Lee thinks there is enormous pressure for facial recognition and other biometric methods to become the norm for verifying one's identity.

> "In my personal view, passwords will be dead in five to ten years' time, in the sense that it will be considered crazy to use them," he said. "Instead, biometrics will become synonymous with the way we transact."

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#Biometrics #Cybersecurity #Deep Technology #Facial Recognition #Privacy

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Shuzhen received her Ph.D. from the Johns Hopkins Bloomberg School of Public Health, USA, where she studied the immune response of mosquito vectors to dengue virus. She subsequently did post-doctoral work on arbovirus evolution at the Genome Institute of Singapore. As a result, she has long since lost count of the number of mosquitoes she has dissected, but harbors a healthy respect for these fascinatingly complex creatures. Now an editor at Asian Scientist Magazine, she hopes to inspire excitement about science through her writing.

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